$$= \frac{1}{2}\int \sqrt{(2x)^2 + 3^2}\, d(2x)$$

$$= \frac{1}{2}\left[\frac{2x}{2}\sqrt{(2x)^2+3^2} + \frac{3^2}{2}\log\left|2x + \sqrt{(2x)^2+3^2}\right|\right] + c$$

$$= \frac{x}{2}\sqrt{4x^2+9} + \frac{9}{4}\log\left|2x + \sqrt{4x^2+9}\right| + c$$

Example 2.54

Evaluate $\int \sqrt{x^2 - 4x + 3}\, dx$

Solution:

By completing the squares

$$x^2 - 4x + 3 = (x-2)^2 - 4 + 3$$
$$= (x-2)^2 - 1^2$$
$$\therefore \sqrt{x^2 - 4x + 3} = \sqrt{(x-2)^2 - 1^2}$$

$$\int \sqrt{x^2 - 4x + 3}\, dx$$

$$= \int \sqrt{(x-2)^2 - 1^2}\, dx$$

$$= \frac{(x-2)}{2}\sqrt{(x-2)^2 - 1^2} - \frac{1}{2}\log\left|(x-2) + \sqrt{(x-2)^2 - 1^2}\right| + c$$

$$= \frac{(x-2)}{2}\sqrt{x^2 - 4x + 3} - \frac{1}{2}\log\left|(x-2) + \sqrt{x^2 - 4x + 3}\right| + c$$

Example 2.55

Evaluate $\int \frac{1}{x - \sqrt{x^2 - 1}}\, dx$

Solution:

By rationalisation,

$$\frac{1}{x - \sqrt{x^2-1}} = \frac{1}{x - \sqrt{x^2-1}} \times \frac{x + \sqrt{x^2-1}}{x + \sqrt{x^2-1}}$$
$$= x + \sqrt{x^2 - 1}$$

$$\int \frac{1}{x - \sqrt{x^2-1}}\, dx$$

$$= \int \left[x + \sqrt{x^2 - 1}\right] dx$$

$$= \int x\, dx + \int \sqrt{x^2 - 1}\, dx$$

$$= \frac{x^2}{2} + \frac{x}{2}\sqrt{x^2-1} - \frac{1}{2}\log\left|x + \sqrt{x^2-1}\right| + c$$

Exercise 2.7

Integrate the following with respect to *x*

1. $\frac{1}{9 - 16x^2}$
2. $\frac{1}{9 - 8x - x^2}$
3. $\frac{1}{2x^2 - 9}$
4. $\frac{1}{x^2 - x - 2}$
5. $\frac{1}{x^2 + 3x + 2}$
6. $\frac{1}{2x^2 + 6x - 8}$
7. $\frac{e^x}{e^{2x} - 9}$
8. $\frac{1}{\sqrt{9x^2 - 7}}$
9. $\frac{1}{\sqrt{x^2 + 6x + 13}}$
10. $\frac{1}{\sqrt{x^2 - 3x + 2}}$
11. $\frac{x^3}{\sqrt{x^8 - 1}}$
12. $\sqrt{1 + x + x^2}$
13. $\sqrt{x^2 - 2}$
14. $\sqrt{4x^2 - 5}$
15. $\sqrt{2x^2 + 4x + 1}$
16. $\frac{1}{x + \sqrt{x^2 - 1}}$

2.2 Definite integrals

So for we learnt about indefinite integrals on elementary algebraic, exponential, trigonometric and logarithmic functions. Now we are going to study about the definite integrals.

Geometrically, definite integral $\int_a^b f(x)\, dx$ represents the limit of a sum. It is also represented as the area bounded the curve $y = f(x)$, the axis of *x*, and the ordinates $x = a$ and $x = b$.

2.2.1 The fundamental theorems of Integral Calculus

Theorem 2.1 First fundamental theorem of Integral Calculus:

If $f(x)$ is a continuous function and $F(x) = \int_a^x f(t)\, dt$, then $F'(x) = f(x)$.

Theorem 2.2 Second fundamental theorem of Integral Calculus:

Let $f(x)$ be a continuous function on [a,b], if $F(x)$ is anti derivative of $f(x)$, then $\int_a^b f(x)\, dx = F(b) - F(a)$.

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